Titles :

JIRONE : *Canteen Automation System*

Bachelor of Technology in Computer Science & Engineering

in

Faculty of Engineering

<u>Submitted by</u>

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Guwahati-26, Assam

Session: January-June, 2024

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DECLARATION

I am, **Raj Dey** bearing **Enrolment ID : ADTU/1/2023-27/BTCS/004** , hereby declare that the project report entitled **"Jirone : Canteen Automation System"** is an original work carried out in the **Department of Computer Science & Engineering**, **Assam Down Town University**, Guwahati.

This work is being submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science & Engineering**. I further declare that this work has not been submitted to any other university or institution for the award of any degree or diploma. The data and findings discussed in this report are the outcome of my own research and development work.

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Semester : 5th

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Date:

(Signature of the Student) **Place: Guwahati**

ACKNOWLEDGMENT

I would like to extend my heartfelt appreciation to everyone who contributed to the successful completion of this project, "**Jirone.**"

First and foremost, I express my deep gratitude to my project supervisor, **Dr.prasenjit k. Das**, for their invaluable guidance, constant encouragement, and constructive feedback throughout the development of this system. Their insights into software architecture and user experience were instrumental in shaping the final outcome.

I is also grateful to the **Faculty of Computer Technology** and the **Department of Computer Science & Engineering** at **Assam Down Town University** for providing me with the necessary infrastructure and resources to carry out this work.

Lastly, I want to thank my friends and family for their patience and encouragement, which helped me stay motivated to persevere through the technical challenges faced during the independent development of this project.

ABSTRACT

This project, titled **"Jirone,"** delves into the realm of automation by developing a dedicated canteen management system for **Assam Down Town University (ADTU)**. The existing manual system for ordering food in campus canteens often leads to overcrowding, long waiting times, and inventory management issues during peak hours.

Jirone addresses these challenges by leveraging a **Serverless Architecture** using **Google Firebase**. The application features a unique "Block-Specific" filtering logic, ensuring students order food available only in their specific building (Block A, B, J, or K). Key technologies implemented include **Cloud Firestore** for real-time database management, **Firebase Authentication** for secure user access, Html5-QRCode for location verification via table scanning, and **EmailJS** for automated digital invoicing.

By enabling students to browse menus, verify their table location, and receive real-time order status updates ("Pending" to "Ready") directly on their smartphones, Jirone significantly reduces queue times and operational inefficiencies. This project demonstrates the practical application of modern web technologies to solve real-world logistical problems within an educational institution

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2	System Architecture
3	Data Flow Diagram
4	Context diagram
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6	Use case Diagram
7	Sequence diagram
8	Schema Diagram
9	Wireframes/Ui

1. INTRODUCTION

1.1 Overview of the project

This project delves into the realm of serverless application development using **Google Firebase** as its primary platform. The application, **Jirone**, leverages various services such as **Firebase Authentication**, **Cloud Firestore**, and **Storage** to provide a scalable and cost-effective solution for canteen management. Through practical implementation, key components such as "Block-Specific" menu filtering, real-time order tracking, and digital invoicing have been developed. This project explores the creation of a fully functional Single Page Application (SPA). By providing a seamless interface for both students and kitchen staff, this project aims to modernize the campus dining experience.

1.2 Motivation

The motivation behind this project stems from the inefficiencies of the traditional manual ordering system at university canteens. With the increasing student population, physical counters often face overcrowding, leading to long wait times and potential errors in order processing. By choosing to focus on a **Serverless Architecture**, this project aims to delve into the core principles of rapid deployment and real-time data handling. Motivated by the opportunity to apply classroom knowledge to a real-world problem at ADTU, this project seeks to equip the university with a digital solution that simplifies logistics and enhances the student experience.

1.3 Scope and Objective

The scope of this project encompasses the exploration and implementation of a web-based ordering system utilizing **Vanilla JavaScript** and **Firebase**. Key areas within this scope include:

- **Student Interface:** Developing features for menu browsing, block selection (A, B, J, K), and QR-based location verification.
- **Admin Interface:** Creating a dashboard for stock management and order status updates.
- **Integration:** Utilizing EmailJS for notifications and HTML5-QRCode for scanning.

Objectives include:

1. Demonstrating the feasibility of a cashless, token-less ordering system.
2. Reducing the average waiting time for students during break hours.

3. Providing kitchen staff with real-time data to manage inventory efficiently.
4. Documenting the development lifecycle of a modern web application.

1.4 Existing system

The existing system encompasses a manual, counter-based approach. In this system, students must physically queue to view a paper menu, make a cash or UPI payment, and receive a paper token. They then wait near the counter until their number is called. However, such architectures often face challenges related to scalability during peak hours. Physical menus are rarely updated in real-time, leading to students ordering out-of-stock items. Additionally, the lack of digital records makes it difficult to track purchase history or manage disputes. These limitations underscore the need for a digital transformation.

1.5 Problem Definition

The problem addressed by this project lies in the inefficiencies and limitations of traditional manual canteen operations. Existing systems struggle to adequately scale and adapt to fluctuating demand during short break intervals, leading to performance decreases (slow service) and increased operational stress. Moreover, the complexity of managing physical cash and tokens detracts from the focus on delivering quality food. As such, there is a pressing need for a more agile and automated solution. By leveraging **Jirone**, this project seeks to address these challenges by providing a flexible, user-friendly platform tailored to the specific infrastructure of ADTU blocks.

1.6 Proposed System

The proposed system for this project centers on leveraging a **Centralized Web Application** to connect students and the kitchen.

- **Block Logic:** The system filters food based on the student's location (Block A, B, J, K), preventing logistical errors.
- **Real-Time Sync:** Using Firestore listeners, order statuses update instantly from "Pending" to "Ready" on the student's device without refreshing.
- **Digital Invoicing:** Automated emails replace paper receipts.
- **QR Verification:** Scanning table QR codes ensures orders are linked to specific seats, enabling "table service" capabilities.

Through comprehensive documentation, the proposed system aims to deliver a robust solution that serves as a model for adopting smart-campus technologies.

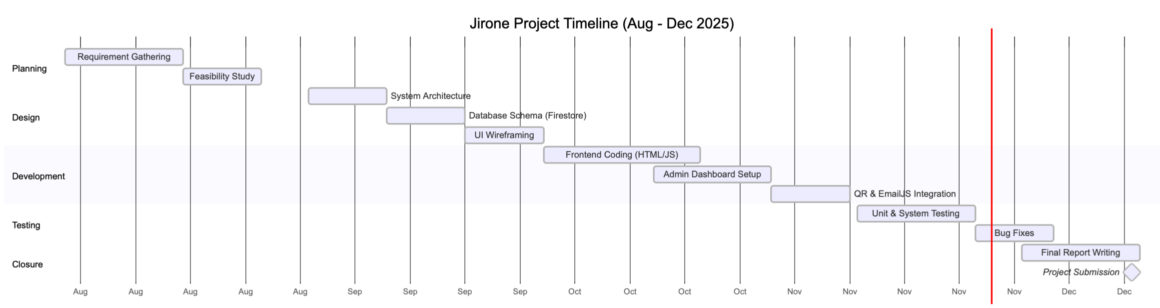
2. PROJECT ANALYSIS

2.1 Project Requirement Analysis

The project entails the development of a serverless application using **Firebase**. The analysis phase involved a thorough examination of stakeholder requirements (Students and Canteen Staff).

- **Functional Requirements:** The system must allow login via Email/Google, filter menus by category (Veg/Non-Veg), and support image uploads for new menu items.
- **Non-Functional Requirements:** The app must load under 3 seconds on 4G networks and ensure data privacy for user contact details.
- **Validation:** Use cases were defined to illustrate user flows, and requirements were detailed to guide the design process.

2.2 Gantt Chart



2.3 Advantage and Disadvantage

Advantage:

1. **Scalability:** Leveraging Firebase allows the system to handle hundreds of concurrent users during break time without crashing.
2. **User Experience:** The "Zomato-like" interface reduces the learning curve for students.
3. **Reduced Operational Overhead:** Automated calculations and digital tokens reduce manual work for staff.

Disadvantages:

4. **Internet Dependency:** The system requires an active internet connection; offline mode is currently limited.
5. **Platform Dependence:** The backend relies heavily on Google's ecosystem, creating a dependency on their pricing and uptime.
6. **Digital Literacy:** Some kitchen staff may require training to use the digital dashboard effectively.

2.4 Project Lifecycle

The project lifecycle involves several key phases:

i) Initiation: Identifying the "Queue Problem" at the ADTU canteen.

ii) Planning: Designing the Block-based filtering logic and UI wireframes.

iii) Execution: Developing the app using VS Code, GitHub, and Firebase Console.

iv) Monitoring and Control: Testing features like QR scanning and adjusting code based on feedback.

v) Testing: Validating orders and stock updates.

vi) Deployment: Hosting the app via GitHub Pages or Firebase Hosting.

vii) Closure: Writing this report and submitting the final project.

2.5 Project feasibility

i) Technical Feasibility: The project is feasible using standard web technologies (HTML/JS) and free-tier cloud services.

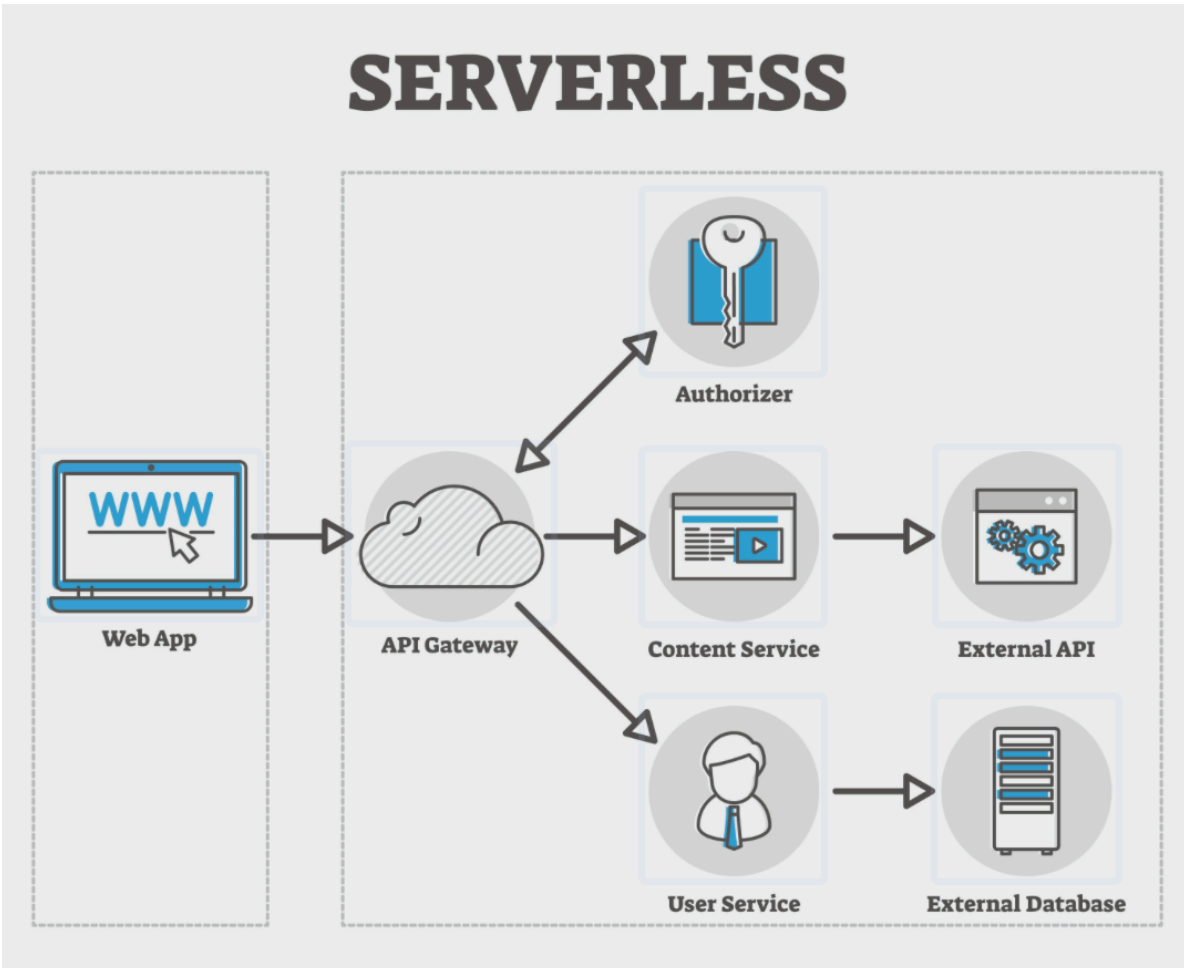
ii) Market Feasibility: There is high demand as every student eats at the canteen; the target market is 100% of the campus population.

iii) Financial Feasibility: Development costs are near zero (open-source tools), making it highly viable.

iv) Operational Feasibility: The UI is designed to be intuitive, requiring minimal training for adoption.

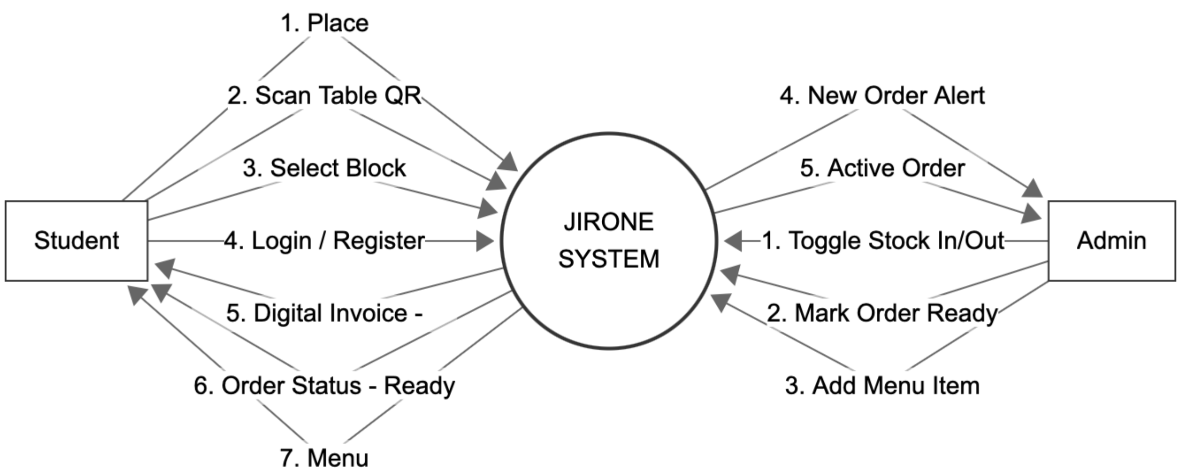
3. PROJECT DESIGN

3.1 System Architecture

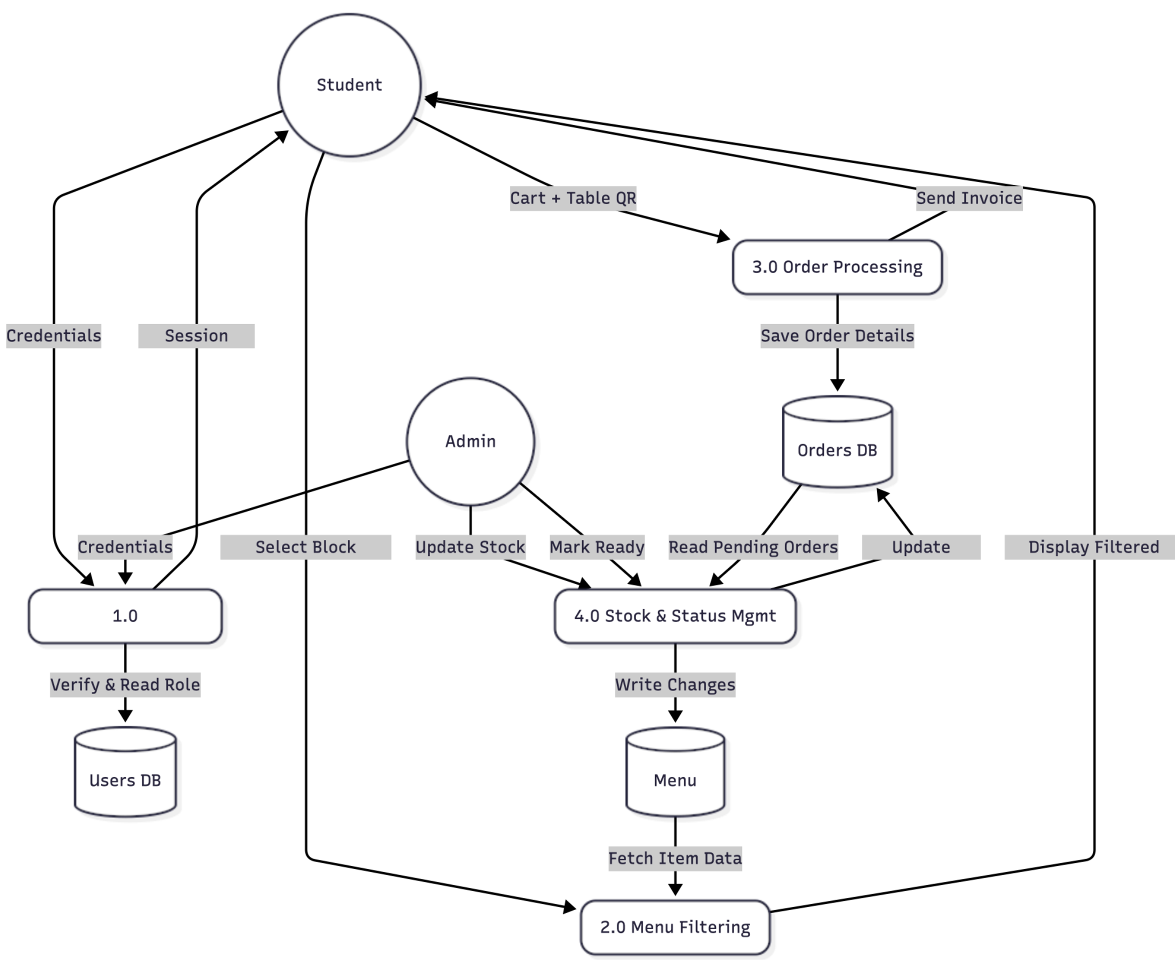


3.2 Data Flow Diagram

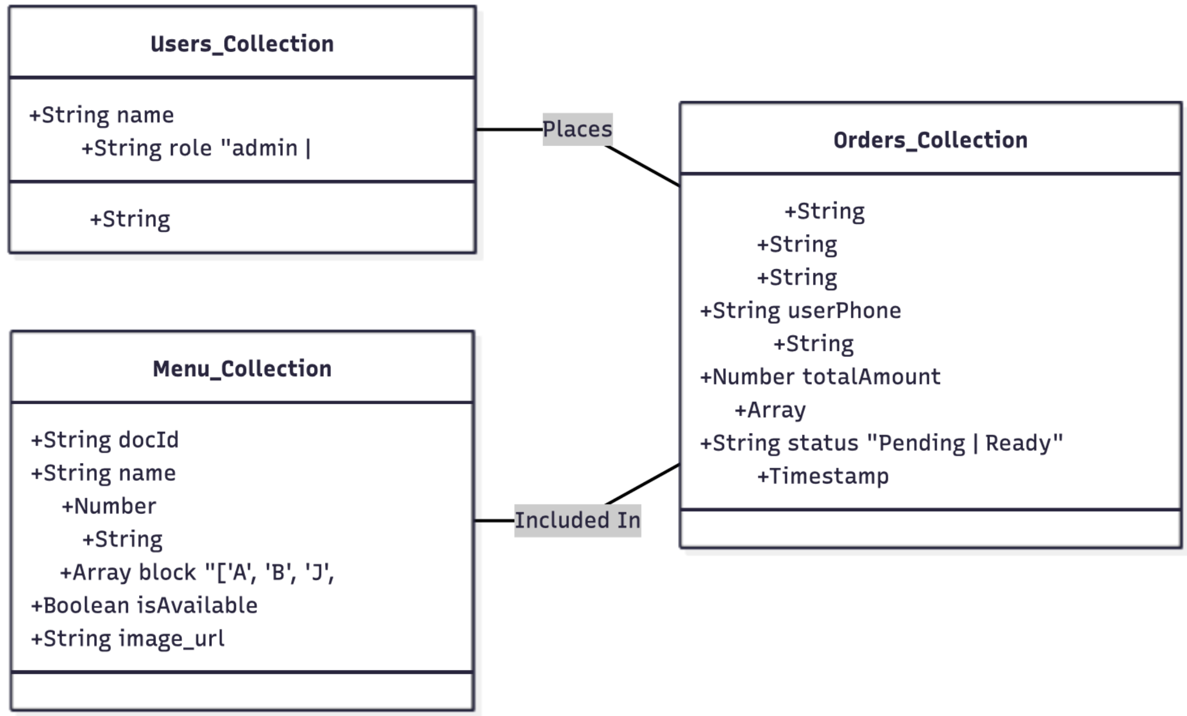
3.2.1 Context Diagram



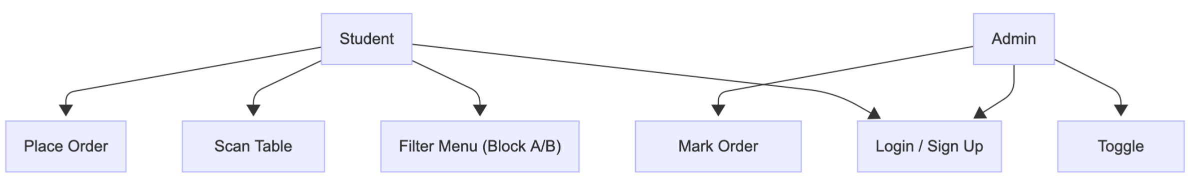
3.2.2 DFD level 1



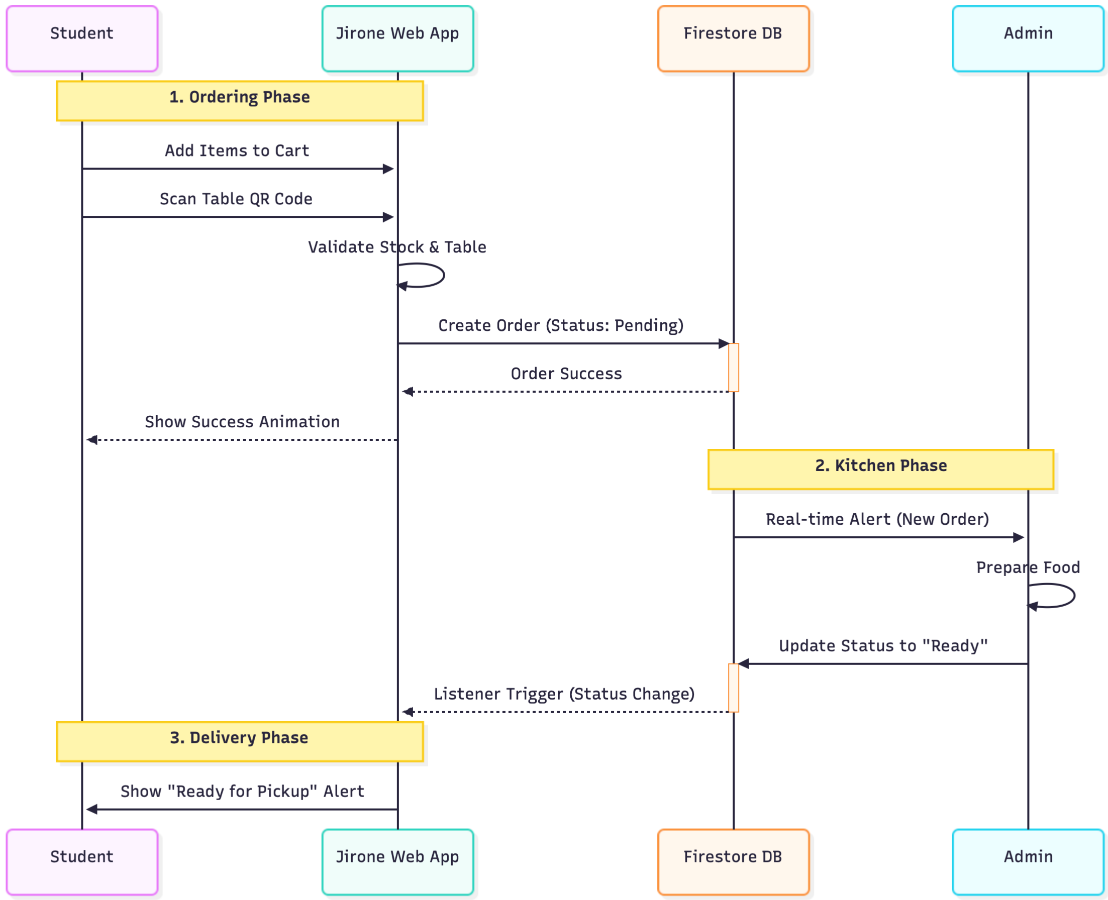
3.3 NoSQL Diagram



3.4 Use case Diagram



3.5 Sequence Diagram



4. PROJECT IMPLEMENTATION

4.1 Description of the software used

I have used **Vanilla JavaScript (ES6 Modules)** for the front-end logic and **Firebase** for the back-end:

i) Jirone Web App (Front-end):

- Developed using standard HTML5 and CSS3 for layout.
- Uses ES6 Modules (import/export) to manage Firebase SDK dependencies.
- Implements a custom State Management system (state object in app.js) to handle cart and user data dynamically.

ii) Cloud Firestore:

- A NoSQL document database used for storing Menu and Order data.
- Real-time listeners (onSnapshot) are used to instantly reflect changes on the UI without reloading.

iii) Firebase Authentication:

- Manages user lifecycle (Sign Up, Login, Logout).
- Supports both Email/Password and Google OAuth.

iv) EmailJS:

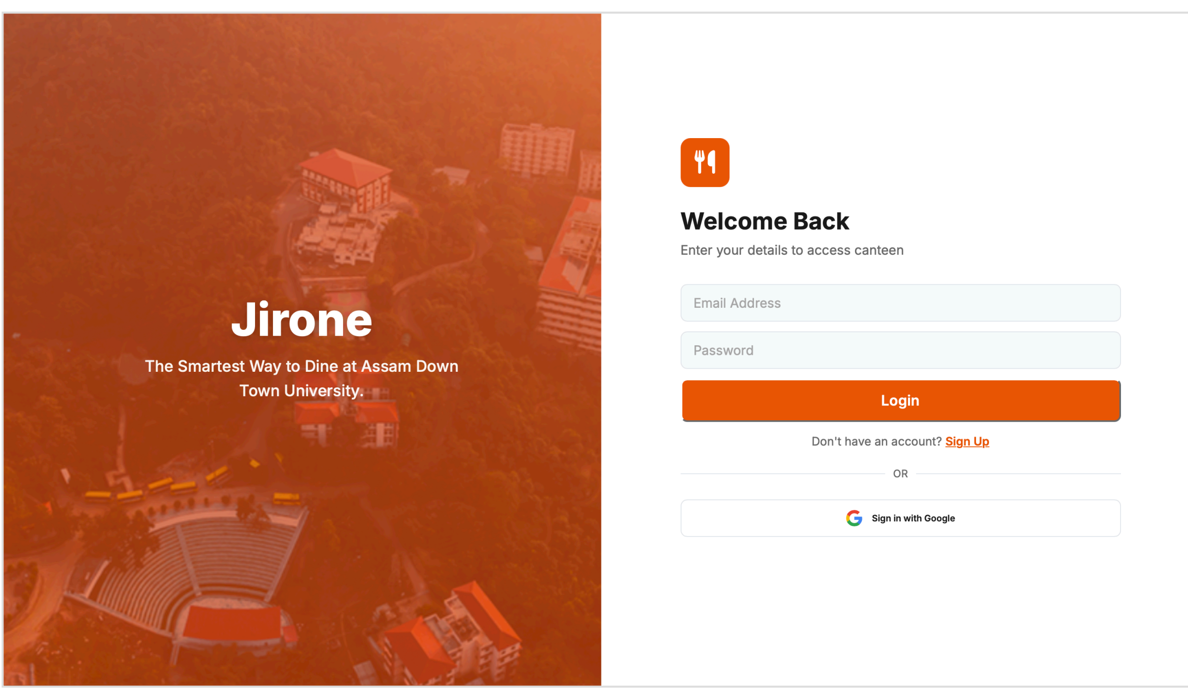
- A client-side email service integrated to send transactional emails (Invoices) directly from the JavaScript code.

v) HTML5-QRCode:

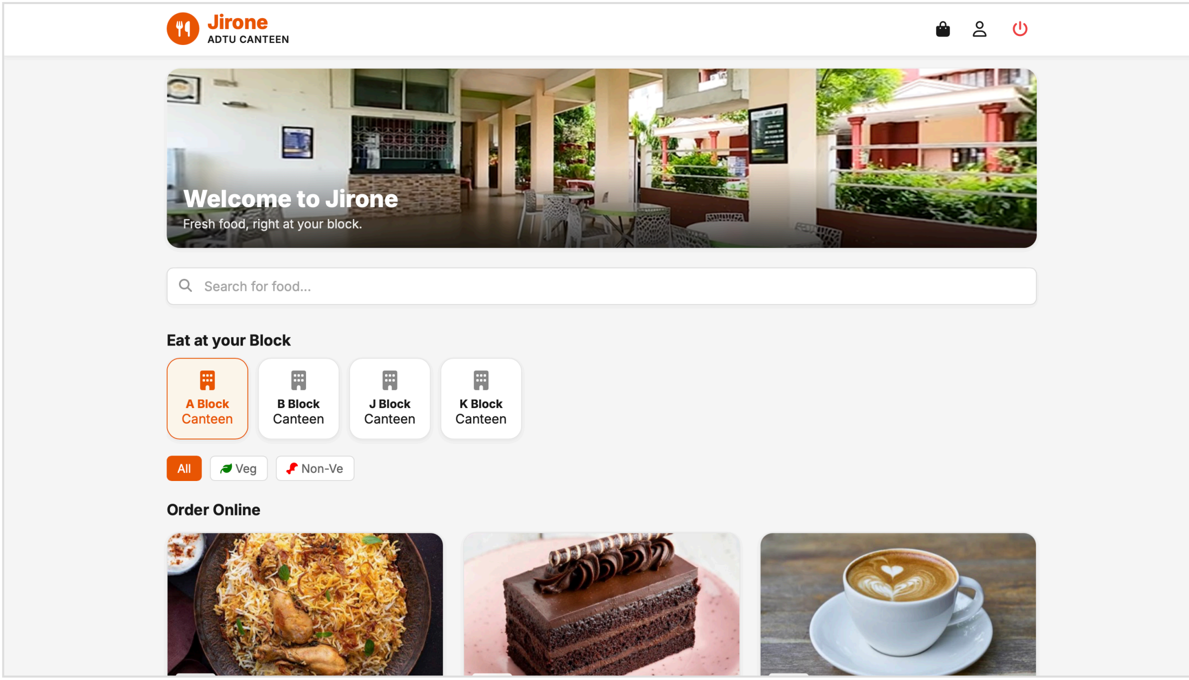
- A lightweight library used to access the device camera and parse QR codes for table identification.

4.2 Wireframes/ Ui

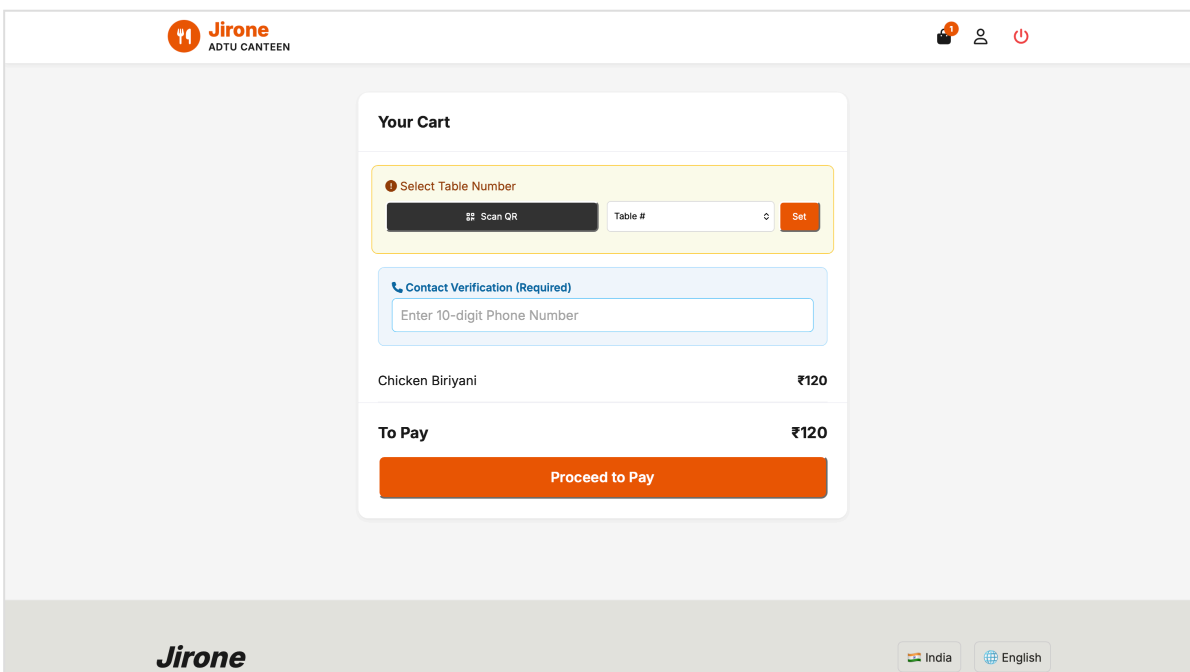
Login Page



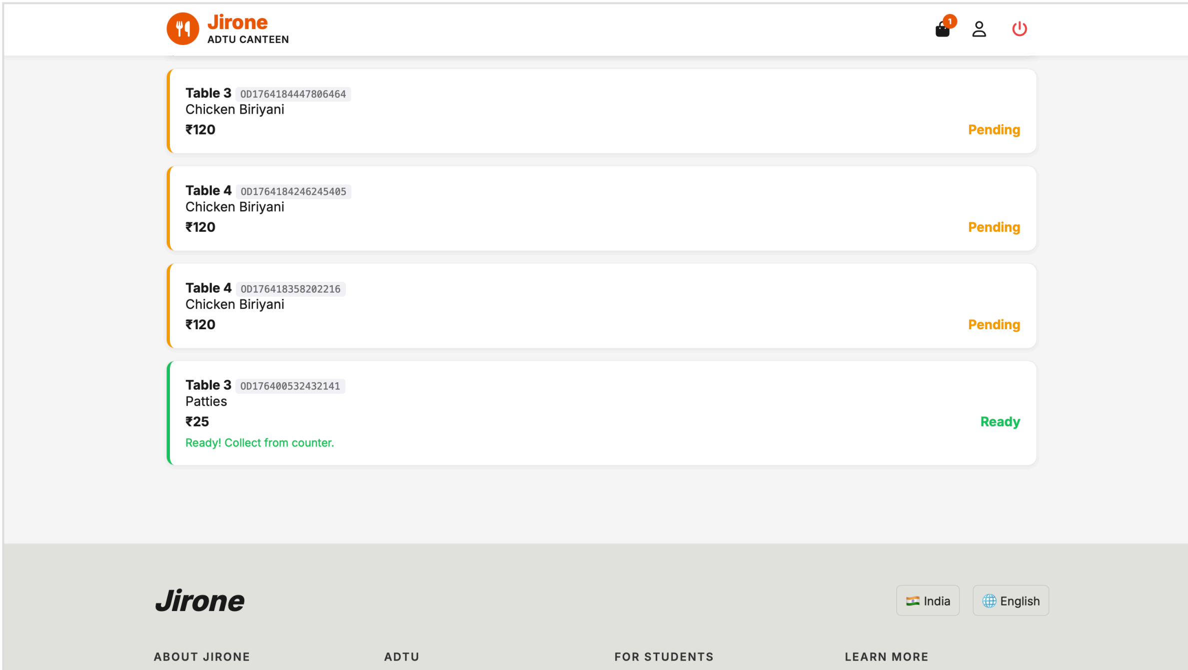
Home Page



Cart

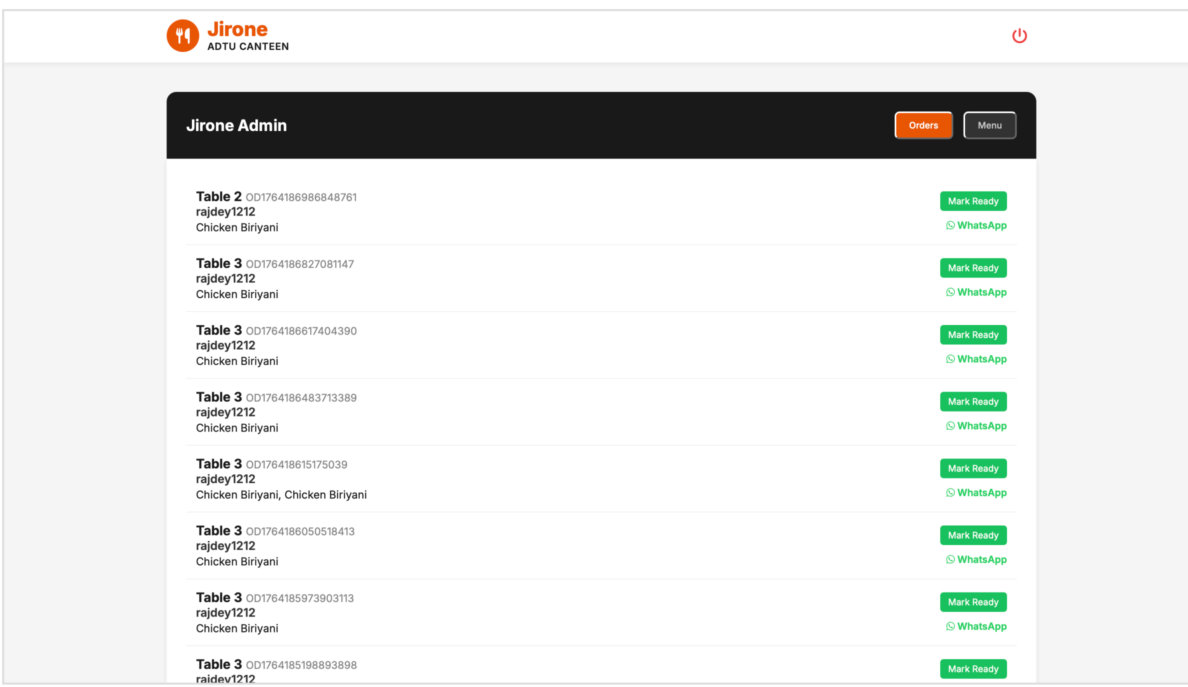


Order History

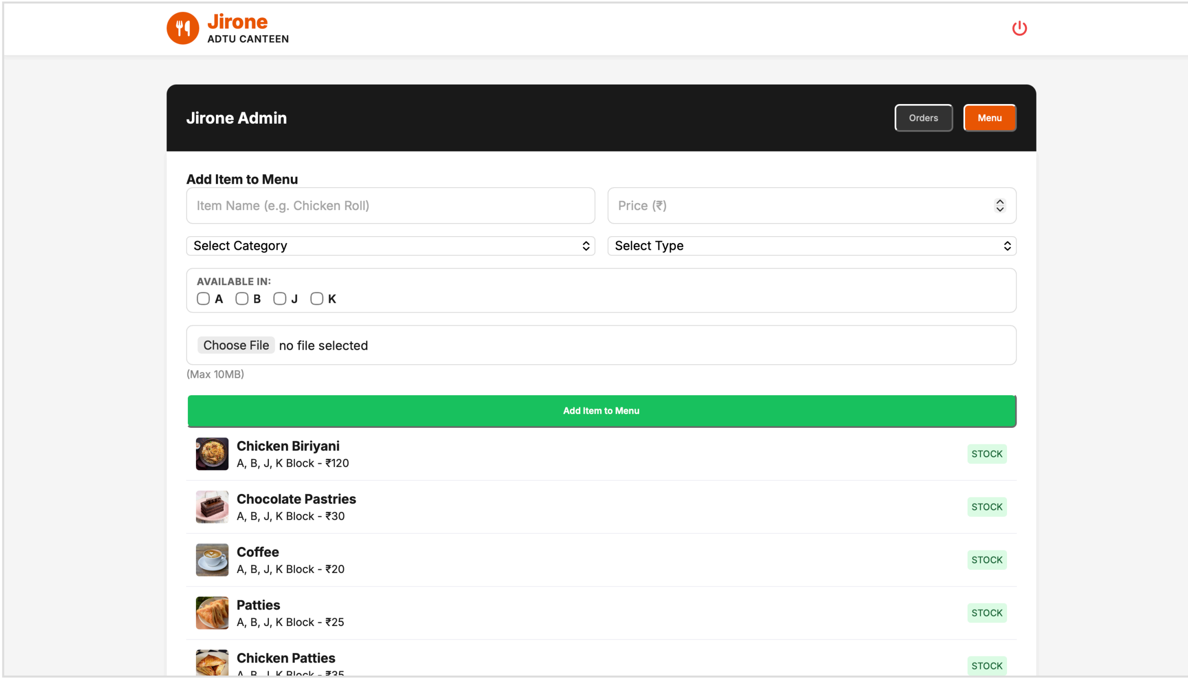


ADMIN Page

Order list



Add Menu (Food)



5. Testing / Result Analysis

5.1 Types of Testing

Some common types of testing used to ensure reliability and functionality:

i) Unit Testing: Tested individual functions like addToCart and handleSearch to ensure they manipulate the local state correctly.

ii) Integration Testing: Verified that the frontend correctly communicates with Firebase Firestore (Database writes) and EmailJS (API calls).

iii) System Testing: Validated the complete workflow: Login -> Order -> Admin Update -> Client Notification.

iv) Compatibility Testing: Checked the application on both Desktop (Chrome) and Mobile (Android Chrome) to ensure responsiveness.

5.3 Test Cases

Test Case	Scenario	Expected Result	Actual Result
TC-01	Login Validation	Enter invalid email format.	System shows "Invalid Email Address" error.
TC-02	Block Filtering	Select "Block A".	Only items tagged with "A" are displayed.
TC-03	Stock Management	Admin toggles item to "Out"	Student sees "SOLD OUT" button immediately.
TC-04	QR Scanning	Click "Place Order" without scan.	Error toast "Scan Table QR First" appears.
TC-05	Order Processing	Place valid order.	Order appears in Admin Dashboard and Email is sent

6.1 Conclusion

The serverless application **Jirone** built using Firebase services offers a modern solution to the age-old problem of canteen queues. By leveraging **Cloud Firestore**, the application achieves real-time synchronization between the kitchen and students, ensuring transparency. The integration of **QR Code technology** for table identification and **Block-specific filtering** demonstrates a high level of customization for the ADTU environment. The implementation of rigorous testing ensures the application is robust and user-friendly. This project successfully meets its objectives of creating a scalable, cost-effective, and efficient canteen automation system.

6.2 References:

- Firebase Documentation: https://firebase.google.com/docs
- EmailJS Integration Guide: https://www.emailjs.com/docs/
- MDN Web Docs: https://developer.mozilla.org/
- Html5-QRCode Library: https://github.com/mebjas/html5-qrcode